

Claude Code for Accounting Research

Module 1 — Foundations: Tools, Context, and the Research
Map

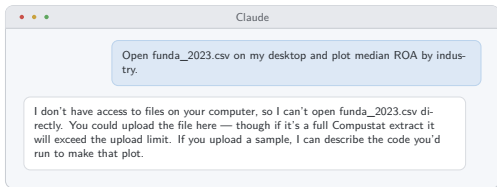
John Barrios

Yale School of Management

Today's Arc

- **0:00–0:40** Lecture: tools, context, verification, research map
- **0:40–1:15** Live demo: one task, start to finish
- **1:15–1:25** Break
- **1:25–2:45** Lab: install, CLAUDE.md, one verified task
- **2:45–3:05** Debrief + exit ticket

“AI Can’t Do Real Research Work”?



Stylized session transcript (not a screenshot).

Chat-level tool: no filesystem, no execution — rung 1 of the ladder.

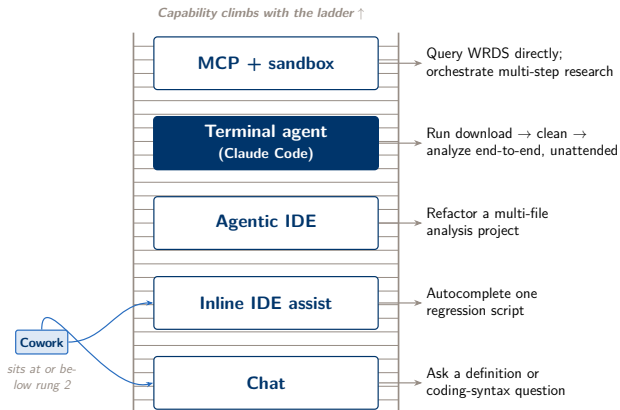


Stylized session transcript (not a screenshot).

Terminal agent: reads files, writes a named script, runs it locally — rung 4.

Same task, different rung. The limit was the tool you tried, not agentic AI.

The Tool Ladder



Claude Code is on rung 4: it works in your terminal, reads your files, and runs code.

Common Failure

Dismissing AI after a Cowork-level disappointment

The tool did what it was built to do, and it was not built to touch your filesystem.
Pick the rung the task needs.

Claude Code as a Terminal-Resident RA

```
demo1 — claude

> Read funda_2023.csv, compute ROA = ni/at for each gvkey-fyear, and plot
median ROA by 2-digit SIC as roa_by_industry.png. Save the script as
roa_plot.py.

• Read(funda_2023.csv)
  ↳ 48,211 rows · gvkey, fyear, ni, at, sic sampled

• Write(roa_plot.py)
  ↳ 41 lines

• Bash(python roa_plot.py)
  ↳ wrote roa_by_industry.png — 12 industries, 2000-2023

Done. One number to verify by hand: median ROA for SIC 28 in 2020 is 0.041.

> Try an edit, or ask a question

~/demo1 · Sonnet · context: 12% used
```

1. your prompt — goal-first, names file, operation, aggregation, output
2. tool calls — the agent acting on your machine
3. results it saw — you can read everything it read
4. handback + a verification hook
5. where you are: directory, model, context used

Stylized session transcript (not a screenshot). Anatomy of a Claude Code turn.

It uses your own R/Python/Stata and your files. Manage it like an RA, not a search engine.

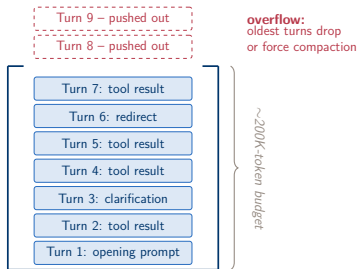
Common Failure

Assuming the runtime is handled for you

It has no Python or R of its own; it uses whatever is installed locally. “Command not found” means check your environment, not the agent.

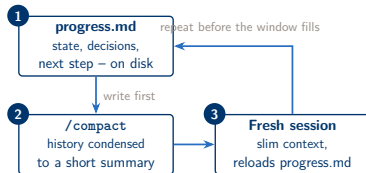
Context Window Mechanics

(a) A finite window, re-sent every turn



Every message re-sends the full history – nothing is remembered independently.

(b) Intentional compaction



Mantra 1 – Files persist; context doesn't.

Write state to disk before it's gone.

Every turn re-sends the whole conversation.

Standard window: 200K tokens (some models: 1M); /context shows how full it is.

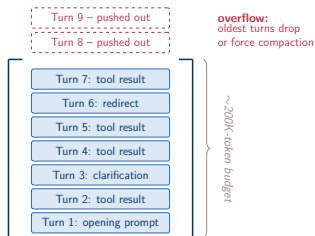
Common Failure

Importing your Dropbox mental model

Nothing persists across sessions unless it was written to a file a future session can read.

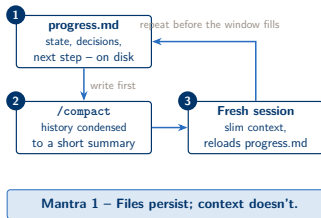
Degradation and Intentional Compaction

(a) A finite window, re-sent every turn



*Every message re-sends the full history –
nothing is remembered independently.*

(b) Intentional compaction



Write state to disk before it's gone.

In long sessions the window fills, and quality can drop before it is full.

Mantra 1

Files persist; context doesn't.

Common Failure

Treating compaction as forgetting everything

Compaction shortens the session; it does not erase it. Write the file, then compact.

Trust, But Verify — The RA Standard

- Delegate freely; check the result before you build on it
- Same standard you apply to a real RA's work
- Pick one case you know well and check it by hand

Verify this

Check one number against a source Claude did not touch before you trust the rest.

Goal-First Prompting: Four Components

- **File** — what you're operating on
- **Operation** — the transformation to perform
- **Aggregation** — the level the result lives at
- **Output** — name and format, in advance

Vague vs. Specific

Vague:

"Look at this Compustat file and tell me something about leverage."

Specific:

"Read funda_1990_2023.csv. For each gvkey-fiscal year, compute book leverage as long-term debt / total assets, winsorize at 1st/99th pctile, mean by fiscal year x 2-digit SIC. Save leverage_by_industry_year.csv and leverage_trends.png."

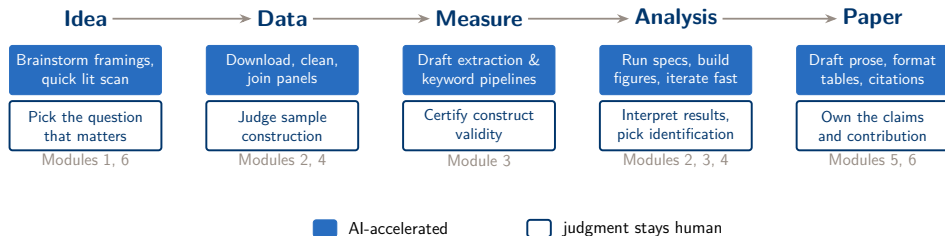
Common Failure

Describing the goal instead of the task

“Analyze this and tell me what’s interesting” hands the agent every decision. Before you hit enter, check for file, operation, aggregation, and output.

The Accounting Research Workflow Map

The accounting research pipeline, annotated by course module



idea → data → measure → analysis → paper
An agent speeds up some stages; the others stay with you.

The Three Module 1 Mantras

Carry these all week

1. Files persist; context doesn't.
2. Trust, but verify — same as an RA.
3. Be specific: file, operation, aggregation, output.

They go on your verification protocol card. Module 6 returns to a harder version of the question.

Next: Live Demo (0:40–1:15)

- One task, start to finish, in a fresh `~/demo1/`
- Launch, permission modes, goal-first prompt (SEC FSDS)
- Watch the loop; redirect, don't argue (Esc)
- Plan Mode preview; `CLAUDE.md`; verify the count

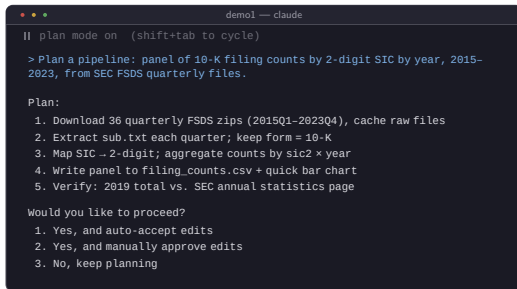
What You'll See: The Permission Ask



Stylized session transcript (not a screenshot).

Manual: asks before every file edit, command, or network call.
Auto: a classifier model passes routine actions, blocks risky ones.
New sessions start in Auto; Shift+Tab to change.

What You'll See: Plan Mode



```
demo1 — claude
II plan mode on (shift+tab to cycle)

> Plan a pipeline: panel of 10-K filing counts by 2-digit SIC by year, 2015-
2023, from SEC FSDS quarterly files.

Plan:
  1. Download 36 quarterly FSDS zips (2015Q1-2023Q4), cache raw files
  2. Extract sub.txt each quarter; keep form = 10-K
  3. Map SIC → 2-digit; aggregate counts by sic2 × year
  4. Write panel to filing_counts.csv + quick bar chart
  5. Verify: 2019 total vs. SEC annual statistics page

Would you like to proceed?
  1. Yes, and auto-accept edits
  2. Yes, and manually approve edits
  3. No, keep planning
```

Stylized session transcript (not a screenshot).

Design before code: review the plan, then execute — or don't.

Shift+Tab cycles Manual → accept-edits → Plan Mode; the status bar shows which.

Next: The Lab (1:25–2:45)

- **Part A (25 min):** environment — `claude` launches, `~/lab1/`
- **Part B (15 min):** write `CLAUDE.md`, restart, confirm it loads
- **Part C (40 min):** tiered task — base / standard / stretch
- Verify checklist: one output checked against an outside source

Debrief, Exit Ticket, and Homework

- Debrief: what broke? Esc or argue? Whose prompt was vaguer than expected?
- Exit ticket: what clicked, what broke, what to verify next time
- **Homework:** confirm your WRDS account + Duo enrollment; test `ssh wrds` before Module 2